

Sparse, Low-Rank, and Deep

A Unified Compressed-Sensing Framework for Music Source Separation

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The Problem in One Sentence

Given only a finished song, recover its hidden tracks, the **vocals** and the **accompaniment**, that were summed together when it was mixed.

This is hard because mixing destroys information: countless track pairs add up to the exact same song. We need a **prior** that says which split is plausible.

Intuition: a song is a picture of sound

- **Spectrogram.** The short-time Fourier transform turns audio into an image: time on the x-axis, pitch on the y-axis, brightness = energy.
- **Vocals are sparse.** A singing voice lights up only a few horizontal harmonic lines per instant, most of the image is dark.
- **Accompaniment is low-rank.** Repeating chords and drums make the background a sum of a few recurring patterns.

Sparse and **low-rank** are exactly the structures that **compressed sensing** knows how to recover, so the same math that reconstructs MRI and photographs from few measurements applies to audio.

The structure each method exploits

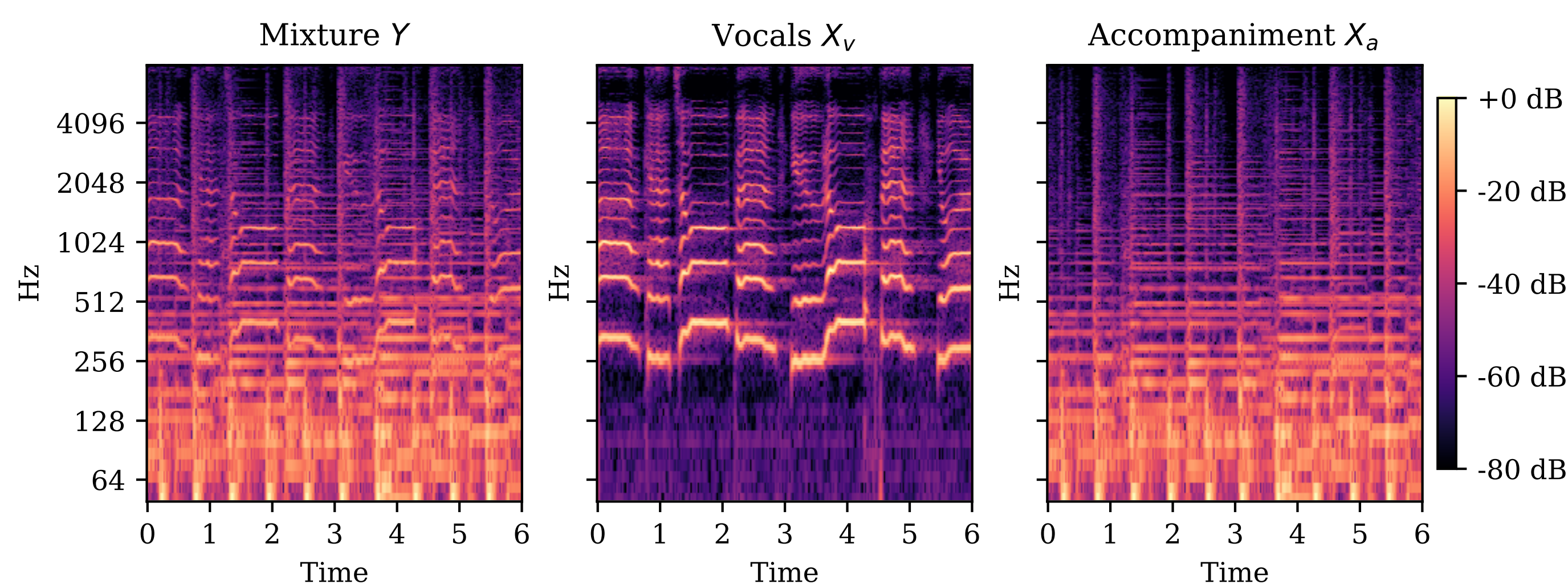


Figure 1: Mixture (left), the vocal track (center, sparse harmonic scaffold), and the accompaniment (right, denser and low-rank). Our methods reverse the sum.

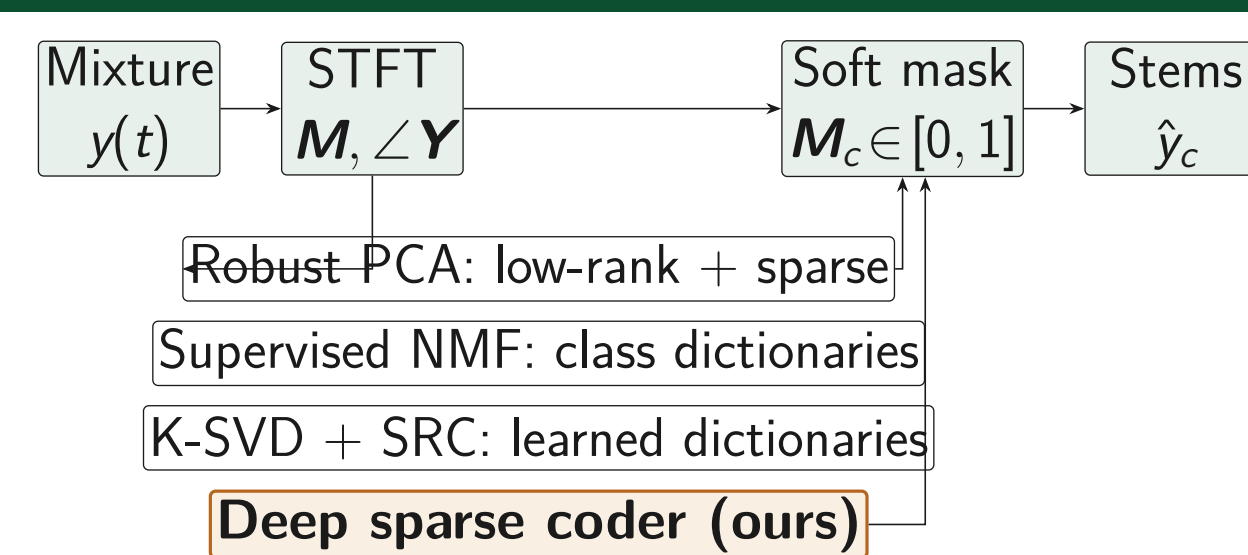
When is recovery guaranteed?

For $\mathbf{y} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}$ s -sparse, the convex program $\min \|\mathbf{z}\|_1$ s.t. $\mathbf{A}\mathbf{z} = \mathbf{y}$ returns $\mathbf{x}$ exactly under *any* of three conditions:

$$\underbrace{\text{spark}(\mathbf{A}) > 2s}_{\text{uniqueness}}, \quad \underbrace{\delta_{2s} < \sqrt{2} - 1}_{\text{RIP}}, \quad \underbrace{s < \frac{1}{2}(1 + \frac{1}{\mu(\mathbf{A})})}_{\text{coherence}}.$$

- **RIP:** $(1 - \delta_s)\|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta_s)\|\mathbf{x}\|_2^2$.
- **Coherence:** $\mu(\mathbf{A}) = \max_{i \neq j} |\langle \mathbf{a}_i, \mathbf{a}_j \rangle|$; random $\mathbf{A}$ needs only $m \geq Cs \log(N/s)$ rows.
- **Low-rank:** nuclear norm $\|\mathbf{X}\|_* = \sum_i \sigma_i$ is the convex surrogate for rank.

One front end, four separators



Every method outputs a mask $\mathbf{M}_c$ that keeps a fraction of each time-frequency cell; the mixture phase is reused to resynthesize audio.

Classical recovery stages

Robust PCA. Split the spectrogram into low-rank + sparse:

$$\min_{\mathbf{X}, \mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \text{ s.t. } \mathbf{X} + \mathbf{E} = \mathbf{M}, \quad \lambda = \frac{1}{\sqrt{\max(F, T)}}.$$

Low-rank $\mathbf{X}$ = accompaniment, sparse $\mathbf{E}$ = voice. No training.

Supervised NMF. Learn a non-negative dictionary per source; KL multiplicative updates $\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{D}^\top (\mathbf{M} \oslash \mathbf{D} \mathbf{H})}{\mathbf{D}^\top \mathbf{1}}$, fix $\mathbf{D} = [\mathbf{D}_v | \mathbf{D}_a]$ at test time.

K-SVD + SRC. Overcomplete dictionaries via rank-one atom updates; route each frame by which source reconstructs it best, $c^* = \arg \min_c \|\mathbf{y} - \mathbf{D}_c \alpha_c\|_2$.

Deep sparse coder (the new method)

Idea. A deep network *is* a sparse-recovery solver with its steps unrolled into layers. We learn the dictionaries by backprop. Each module stacks a *fat* (upsampling) then *tall* (downsampling) non-negative code:

$$\alpha^{(1)} = \arg \min_{\alpha \geq 0} \frac{1}{2} \|\mathbf{x} - \mathbf{D}^{(1)} \alpha\|_2^2 + \lambda_1 \|\alpha\|_1 + \frac{\lambda_2}{2} \|\alpha\|_2^2,$$

solved by K unrolled proximal-gradient steps (all matmul + clamp, runs on GPU):

$$\alpha_k = \left[\mathbf{z}_k - \frac{1}{L} \mathbf{D}^\top (\mathbf{D} \mathbf{z}_k - \mathbf{x}) - \frac{\lambda_1}{L} \right]_+.$$

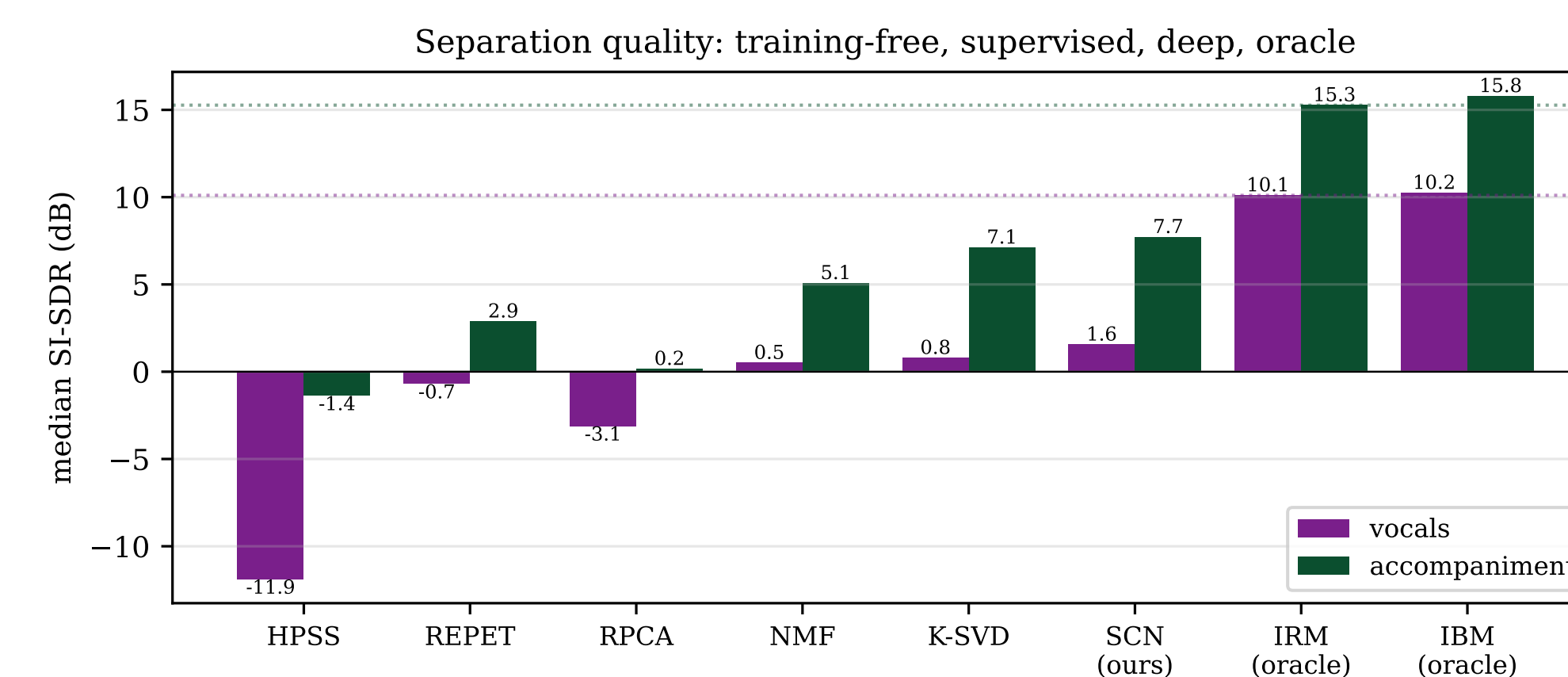
Training. Stack 3 modules + sigmoid; minimize $\|(\mathbf{M}_v - \mathbf{M}_v^{\text{IRM}}) \oslash \mathbf{M}\|_1 + \beta \sum_\ell \|\alpha^{(\ell)}\|_1$. This is LISTA, specialized to non-negative codes and a masking output.

Stereo as joint sparsity

Two channels, one support. Left and right share the same active sources, so their codes share *rows*. Simultaneous OMP scores atoms by the cross-channel row norm $\|(\mathbf{D}^\top \mathbf{R})_{n,:}\|_2$, and the shared support *improves* identifiability: $s < \frac{1}{2}(\text{spark}(\mathbf{A}) - 1 + \text{rank}(\mathbf{Y}))$.

Results on 25 MUSDB18 test tracks

Method	SI-SDR (dB)		Time (s)
	Vocals	Accomp.	
HPSS (median)	−11.9	−1.4	0.25
REPET-SIM	−0.7	2.9	0.08
Robust PCA	−3.1	0.2	1.14
Supervised NMF	0.5	5.1	0.46
K-SVD + SRC	0.8	7.1	0.28
Deep coder (ours)	1.6	7.7	0.08
Oracle IRM / IBM	10.2	15.8	—



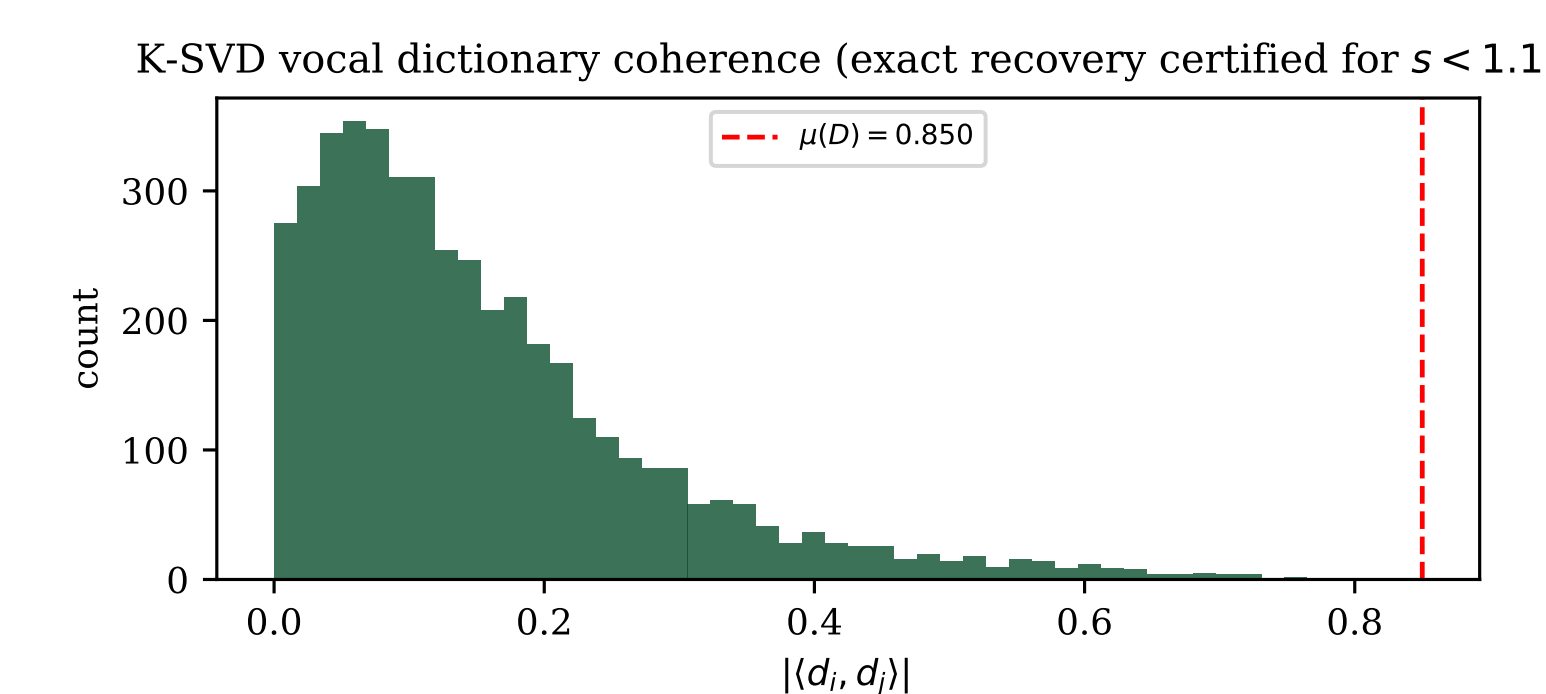
- **Clear ordering:** training-free < supervised < deep.
- **Best of both:** the deep coder wins on both sources *and* is the fastest learned model. Soft masks add +1 dB over hard masks.

What the knobs do (ablations)

NMF atoms K	16	32	64	128
Vocal SI-SDR	2.2	1.9	1.8	1.5
K-SVD s	3	5	10	15
Vocal SI-SDR	1.4	2.3	2.7	2.7
RPCA λ scale	0.5	1.0	1.5	2.0
Vocal SI-SDR	−3.9	−3.6	−3.2	−2.3

Smaller dictionaries generalize better (fewer atoms = stronger prior); **richer codes help** up to $s \approx 10$, then saturate.

Honest negative result: coherence



Learned dictionaries are **coherent** ($\mu \approx 0.85$), so the worst-case theorem certifies recovery only for $s \lesssim 1$, yet $s = 10$ works in practice. **Average-case** $\neq$ **worst-case** for trained dictionaries.

Takeaways

- **Unification.** One front end places low-rank, NMF, dictionary, and deep methods on a single structure-recovery axis.
- **Deep wins.** The unrolled sparse coder is the most accurate non-oracle separator *and* the fastest learned model.
- **More structure-learning** $\Rightarrow$ **better separation**, monotonically.
- **Code, models, audio:** github.com/takakhoo/ENGS109_Final_Project